

Where Square stands, September 2026

Square, wienerlabs/square. Generated from docs/strategy/market-assessment-2026-09.md

A market assessment and a roadmap, written on 17 September 2026, the day after Arc mainnet launched. It answers two questions: does this project make sense, and what is missing. Every figure in it is sourced at the end.

What moved

Arc mainnet went live on 16 September 2026. Its founding validators are BlackRock, DTCC, ICE, Mastercard, Visa, Standard Chartered, MoneyGram, SBI, Sumitomo, Galaxy and Worldpay. Gas is USDC. Circle describes Arc as "the first blockchain designed from genesis for AI agents as economic actors" and shipped Agent Stack the same day: Agent Wallets, an Agent Marketplace, Nanopayments, Circle CLI and Circle Skills. Testnet did not close. Mainnet is chain 1243, testnet remains 5042002, and a hardcoded chain id now needs a second branch rather than an edited value.

In agent payments the narrative is growing faster than the usage. x402 has 165 million transactions and 69,000 active agents, but on-chain data puts real commerce at roughly 28,000 dollars a day, with about half of activity classified as gamified. Through late 2025 the speculative share rose from under 6 per cent in October to 67 per cent in December. The infrastructure arrived first; the economy it is meant to carry has not.

Regulation is moving toward Square rather than away from it. Under the GENIUS Act, FinCEN and OFAC published a proposed rule on 8 April 2026 treating permitted payment stablecoin issuers as financial institutions under the Bank Secrecy Act, with customer identification, anti money laundering and sanctions compliance programmes required. The industry conversation is shifting from know your customer to know your agent: verifiable identities for financial bots, tied to legal entities.

And the layer Square writes its reputation records into has been measured, and it is mostly empty. An empirical study of the ERC-8004 ecosystem found that of 32,343 agents registered on Ethereum only 3 per cent expose a valid registration file; that 98.7 per cent of feedback on Ethereum carries no payment proof and no task linkage, 100 per cent on BNB Smart Chain and 99.3 per cent on Base; that the median record costs 0.0027 dollars to write, against a median agent payment volume 259 times larger; and that on Base 90.6 per cent of reviewers share funding provenance with other reviewers.

Does the project make sense

Yes, and more so than a week ago. Three reasons, each measurable.

The architecture already occupies what the literature calls future work. "Compliance-Aware Agentic Payments on Stablecoin Rails" proposes exactly Square's shape: a policy registry, escrow, dispute resolution. It does not use zero knowledge proofs, relying on deterministic on-chain checks, and it lists privacy-preserving compliance verification as an open problem. Square's `ComplianceModule`, its Groth16 verifier and its Poseidon commitment are an answer to that open problem.

The evidence gap is real and Square closes it structurally. The study's third finding is that ratings carry no verifiable interaction. Square's hook writes its validation record inside `complete`, in the transaction that credits the payee, and binds it to the job through the validation request opened at submit. The study's fourth finding is that a record costs fractions of a cent to forge; in Square, producing one requires funding a job, waiting out the challenge window and settling it. That is a property of the contract, not a claim in a brochure.

Two of Arc's own four stated frontiers describe Square. Arc's request for builders names outcome marketplaces under the agentic economy, and bounded agents with risk limits under intelligent accounts. `PolicyRegistry.dailyLimit` together with `spentToday` is the second one exactly.

Three things should be said plainly against that.

Competition now comes from the chain's operator. Circle Agent Stack ships agent wallets, an agent marketplace and nanopayments. Square has no business in that territory and should not enter it; Square's territory is escrow, dispute and mandate, none of which Agent Stack addresses.

There is an incumbent on the same primitive. Virtuals Protocol's Agent Commerce Protocol is the principal production deployment of ERC-8183, live on Base and Arbitrum, with 12.3 million commerce memos. Square has no users. The difference is the compliance gate, bonded arbitration and the receivable market, but a difference is only a difference once somebody uses it.

And the hardest of the three: **no compliance claim can be made today, because the ceremony has not been held.** The README says so itself, in the words that a single machine could forge a proof for any statement. That is honesty rather than a defect, but it is also the one thing standing in front of an institutional conversation.

What is missing, in order

#	Gap	Why it matters	Where it is written
1	The phase 2 ceremony has not been held (#16)	The whole compliance claim rests on this key and is void without it	<code>docs/disclosure/zk-setup-status.md</code> , <code>docs/ceremony/README.md</code>
2	The validation record did not carry the payment	The evidence claim was half built: task linkage without payment proof	<code>contracts/src/SquareHook.sol</code> , closed by #405
3	No user has ever used it	Against 12.3 million memos elsewhere, Square has demonstration transactions	<code>docs/deploy/lifecycle-5042002-2026-09-09.md</code>
4	The shared stack is 21 selectors behind and no address is verified	Deploying the app or the keeper from <code>main</code> against it would revert	<code>docs/deploy/README.md</code> section 5
5	No service is hosted	Railway is decided, the project does not exist, nothing runs continuously	<code>docs/decisions/service-hosting.md</code>
6	Nothing is published to npm	The only installation path is clone and build	<code>docs/decisions/distribution-channel.md</code>
7	The circuit's fifth rule binds nothing	<code>payment_category</code> is the operator's own statement with nothing on chain to check it against	<code>circuits/README.md</code>
8	Privacy rests on one operator-held salt	A weak <code>policy_salt</code> recovered a 25 USDC ceiling in 50 tries and 10 ms	<code>circuits/README.md</code>

Items 7 and 8 are already written down in the repository. That is the right place for them: a reader who learns a limitation from the project has been told something, rather than having caught it.

The roadmap

Four goals, in one order. They are not parallel, and the order follows from one observation:

The ceremony has the longest lead time and needs no code, while the evidence claim can be made in full without it. So the ceremony starts now and runs in the background, the product reaches a network on the strength of its evidence, and the grant application carries the proof of both.

A. Evidence, the new headline claim

The claim is: every reputation record Square writes has been paid for. The record exists because money moved, and forging one costs the job it names.

- **A1, done in #405.** SquareHook's `responseHash` commits to the settlement: the job, the payee, the amount the kernel credited, the token, the screening commitment and the two check outcomes. A record is now written for every settled job rather than only for gated ones, the response follows the money, and the tag moved to `square.settlement`.
- **A2, done in #405.** The preimage is emitted as `EvidenceRecorded`, because a commitment nobody can recompute is not evidence.
- **A3.** The indexer gains an endpoint serving an agent's payment-backed record set, with amounts, transaction hashes and preimages. `/jobs/provider/:address` exists but is a job list, not an evidence feed.
- **A4, done in #405.** `npm run check:evidence` reads a job's event, the registry's stored hash and the kernel's own settlement facts, and requires all three to agree.
- **A5, done in #405.** `docs/design/evidence-records.md` defines the tag, the commitment and the answer to each of the study's four findings.
- **A6.** The README and the site lead with the claim.

B. Arc mainnet, chain 1243

- **B1.** Redeploy the testnet first (#324). It closes the 21 selector gap and rehearses the mainnet run.
- **B2.** `packages/core/src/deployments.ts` gains a second branch: a chain id, a network profile and a deployment record. The file is already keyed by chain id, so this is an addition rather than an edit, and the testnet entries stay because testnet stays.
- **B3.** Mainnet launches with the compliance gate off. Escrow, the challenge window, bonded arbitration, the receivable market and the evidence records all work without it, and turning it on before the ceremony would contradict the repository's own disclosure.
- **B4.** Ownership moves to a Safe, which the README already requires before mainnet.

C. The ceremony, starting now

Coordination rather than code, and the longest lead time of anything here. The checklist in `docs/ceremony/README.md` has open boxes: participant count, invitation list, at least one institutional participant, window open and close, beacon round, publication location. The announcement is drafted with four bracketed values. Arc's new institutional ecosystem is a real source for that invitation list.

D. Grant and standards

- **D1.** Arc Developer Grants and the Builders Fund are strongest once A and B have evidence behind them: an outcome marketplace running on a network, payment-backed reputation records, and a ceremony with a date.
- **D2.** Two standards contributions. The evidence record written up as an ERC-8004 profile, which answers the study's canonical tag and verifiable interaction gaps directly. And the compliance gate described as an ERC-8183 extension, where the academic gap gives a ready frame.

Sources

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